CLO Analysis

The Assignment

An SPV assembles a pool of 80 identical loans, each one having the following characteristics:

Principal: $1.25m
Maturity: 1 year
Coupon: 6.80%
Payments: Act/Act annually
Credit rating: BB

The SPV estimates that the individual default probability of each loan is 1.2%, consistent with their BB rating. It also estimates that, in the case of a default, there will be a 40% recovery of principal, but no recovery of interest.

The SPV creates a CLO with two tranches as follows:

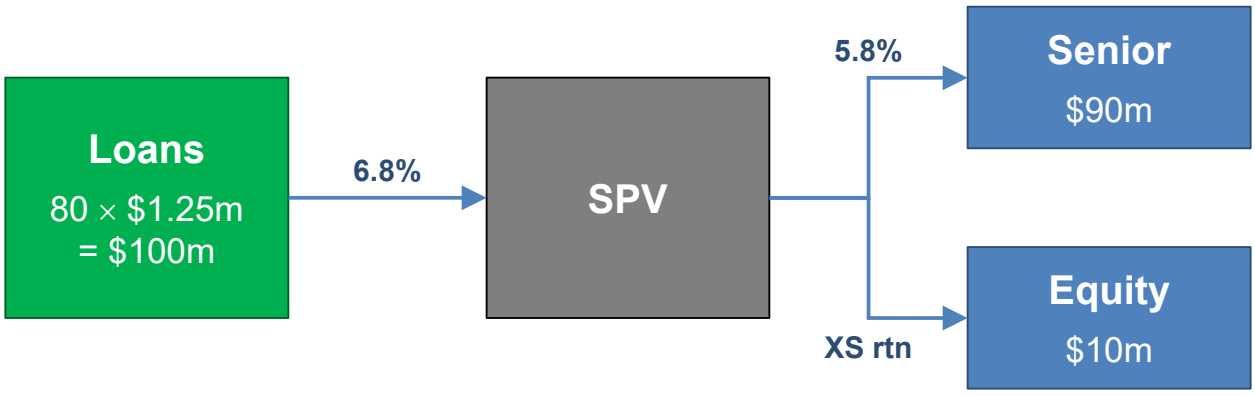
Senior Tranche

Principal: $90m
Coupon: 5.80%
Payments: Act/Act annually
Redemption: Par

Equity Tranche

Principal: $10m
Coupon: Excess return (may be negative)
Redemption: Par (but may be diminished if excess return is negative)

The cash flows are pictured in the diagram below:



© 2002-22 ACF Consultants Ltd
All rights reserved

REQUIRED

1. What is the degree of over-collateralization for holders of the senior tranche?
2. Suppose the minimum acceptable over-collateralization ratio was 108%. What is the maximum number of defaults that can occur *before* the OC ratio is breached?
3. What is the interest-coverage ratio for the senior tranche? (This is the ratio of interest received by the SPV compared to interest paid to the senior tranche.)
4. What is the total amount returned to holders of the equity tranche if there are no defaults?
5. What is the total amount returned to holders of the equity tranche if there is one default? *HINT: The easiest way to determine this is to calculate the financial impact of a single default. This will also help you answer then next two questions.*
6. What is the total amount returned to holders of the equity tranche if there are two defaults?
7. What is the maximum number of defaults that can be sustained *before* holders of the senior tranche begin to suffer?
8. If the incidence of defaults were uncorrelated, what is the chance that holders of the senior tranche will suffer income or capital impairment? *HINT: Try using Excel's BINOMDIST function.*
9. What if the incidence of defaults were positively correlated? What would happen to your answer for the previous question?

© 2002-22 ACF Consultants Ltd
All rights reserved